

REIPA, Norway

WMO: 011140

Lat: 66.9033N

Lon: 13.6458E

Elev: 8

StdP: 101.23

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-11.2	-17.6	0.8	-10.8	-16.0	0.9	-9.8	18.1	5.4	15.1	5.4	2.1	60	0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	22.8	15.6	20.7	14.7	18.9	13.9	16.9	21.7	15.5	19.4	14.5	17.9	2.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.7	10.4	19.0	13.7	9.8	17.3	12.7	9.2	16.2	47.5	21.8	43.5	19.6	40.7	18.0	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.2	10.3	-15.0	25.7	1.6	2.2	-16.2	27.3	-17.1	28.6	-18.1	29.8	-19.2	31.4		
(5)			-15.4	18.5	2.0	1.6	-16.9	19.6	-18.1	20.6	-19.2	21.4	-20.7	22.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-2.1	-1.7	-0.3	2.9	7.3	10.1	13.4	12.9	10.0	5.2	1.7	-0.7
	DBStd	6.59	4.58	4.53	3.97	3.04	3.44	3.01	2.97	2.50	2.75	3.51	4.17	4.46
	HDD10.0	2139	375	328	319	213	99	34	1	4	33	153	248	331
	HDD18.3	4894	633	561	578	462	342	248	157	169	249	407	498	589
	CDD10.0	293	0	0	0	1	16	37	108	94	33	4	0	0
	CDD18.3	7	0	0	0	0	0	1	5	1	0	0	0	0
	CDH23.3	57	0	0	0	0	1	5	46	6	0	0	0	0
	CDH26.7	7	0	0	0	0	0	0	7	0	0	0	0	0
Wind	WSAvg	3.7	3.6	3.9	4.4	3.6	3.6	3.7	3.2	3.1	3.3	3.8	3.8	4.2
Precipitation	PrecAvg	1692	151	133	127	104	102	101	110	133	192	214	175	178
	PrecMax	2068	364	238	216	185	151	207	228	354	363	396	340	328
	PrecMin	1360	25	72	42	25	39	22	29	36	49	60	24	22
	PrecStd	190	87	47	53	41	37	47	53	65	80	88	85	81
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	7.6	9.3	17.0	22.3	23.6	27.6	24.0	19.6	16.4	11.1	9.0
		MCWB	6.2	5.3	4.4	9.0	13.0	14.2	18.5	16.8	12.8	11.1	8.4	5.8
	2%	DB	6.5	6.6	7.2	12.0	19.1	19.8	24.3	21.6	17.8	13.3	9.5	7.8
		MCWB	4.5	4.7	4.6	6.8	11.8	13.3	17.2	15.7	12.2	9.2	6.9	6.0
	5%	DB	5.3	5.8	6.3	8.9	15.9	17.7	21.1	20.1	16.1	11.6	8.6	6.7
		MCWB	3.6	4.0	4.1	5.0	10.6	12.4	15.4	14.9	11.6	8.7	6.4	4.8
	10%	DB	4.2	5.0	5.4	7.5	13.2	15.6	18.9	18.2	14.6	10.2	7.5	5.6
		MCWB	2.4	3.2	3.3	4.4	8.9	11.7	14.9	14.0	11.3	7.9	5.5	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	6.2	6.2	9.6	13.5	15.4	19.6	17.9	14.3	11.6	9.0	7.4
		MCDB	7.6	7.0	7.1	16.1	20.4	20.9	25.5	21.9	17.6	16.0	9.8	8.1
	2%	WB	5.0	5.2	5.3	7.2	12.3	14.2	17.7	16.4	13.1	10.1	7.7	6.5
		MCDB	6.0	6.2	6.7	10.5	18.1	19.1	23.1	20.8	15.8	12.4	9.1	7.4
	5%	WB	3.8	4.4	4.4	5.9	10.8	12.9	16.3	15.2	12.4	9.1	6.8	4.8
		MCDB	5.2	5.7	6.2	7.8	15.1	16.8	20.4	19.1	14.9	11.1	8.3	6.6
	10%	WB	2.7	3.2	3.4	5.1	9.3	11.9	15.0	14.4	11.7	8.1	5.6	3.6
		MCDB	4.2	4.7	5.3	7.1	12.9	14.9	18.2	17.6	14.1	10.0	7.2	5.3
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.3	7.1	7.5	7.3	6.1	6.9	7.6	7.3	5.9	5.4	5.7
		MCDBR	5.5	5.8	6.2	9.9	12.4	10.9	12.2	11.3	9.5	7.9	5.9	5.7
		MCWBR	4.6	5.0	4.7	6.5	6.6	5.9	6.0	6.2	5.6	5.2	4.4	4.9
	5% WB	MCDBR	5.3	5.0	5.3	7.9	11.6	9.1	10.6	10.3	8.1	6.2	5.3	5.4
		MCWBR	4.9	4.8	4.6	5.8	6.6	5.1	5.6	6.1	5.2	4.6	4.7	4.9
Clear-Sky Solar Irradiance	taub	0.368	0.290	0.321	0.343	0.347	0.344	0.352	0.340	0.327	0.298	0.350	N/A	
	taud	2.590	2.166	2.258	2.343	2.418	2.479	2.491	2.515	2.522	2.306	2.541	N/A	
	Ebn at Noon	181	534	725	813	852	864	844	817	739	529	162	0	
	Edh at Noon	17	50	78	93	96	93	89	79	62	43	15	0	
All-Sky Solar Radiation	RadAvg	0.05	0.38	1.52	3.19	4.60	5.15	4.61	3.45	1.85	0.66	0.11	0.00	
	RadStd	0.01	0.07	0.21	0.31	0.32	0.66	0.48	0.34	0.21	0.13	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page